Francisco Pinto Santos

Software Engineer | Backend, Distributed Systems | PhD



- **Location:** Madrid, Spain (remote)
- **Phone:** +34 636 627 009
- **Email:** franpintosantos@gmail.com
- **LinkedIn:** https://linkedin.com/in/franpintosantos
- **GitHub:** https://github.com/gandalfran

About Me

Software Engineer with 7+ years of experience building scalable telco and SaaS products that process high-volume production traffic and optimize daily operations. Highly adaptable and a fast learner, equally comfortable jumping into backend, integrations, identity, distributed data and cloud-native platforms. Backed by a PhD in Computer Science and Artificial Intelligence, combining deep technical foundations with a hands-on, get-things-done mindset.

Skills

Backend: Java, Spring Boot, Node.js, Express.js, TypeScript, Python, Flask, Django, FastAPI, REST APIs, Open Gateway and CAMARA.
Data: Apache Kafka, Apache Spark, SQL, PostgreSQL, MySQL, MongoDB, Elasticsearch, Neo4j, microservices, distributed systems and data-driven architectures.
Platform: Bash, Git, GitHub Actions, CI/CD, Docker, Kubernetes, AKS, Helm, Helmfile, AWS, Azure, OpenFeature, OpenTelemetry, Grafana, Loki, Mimir, Tempo and Prometheus.
Identity: OAuth 2.0, OpenID Connect, CIBA, JWT, JWS, JWE, JWK/JWKS, mTLS and Keycloak.

Languages

Spanish: native.
English: full professional proficiency.

Experience

Software Engineer - Telefónica Innovación Digital
01/2024 - present · Madrid, Spain

- Took ownership of Transparency Center, a React and TypeScript/Node.js component of a distributed consent-management platform handling peak traffic of 10K requests/minute, and contributed to the migration to a multi-tenant model serving +20 tenants.
- Designed and validated REST API integrations and OAuth flows for +30 partners across +40 CAMARA APIs and +20 OAuth and proprietary flows, backed by test collections and public documentation.
- Owned identity tracks across API Gateway and Authentication Server, contributing to +10 consent and authorization flows alongside IdP integrations and strengthened authentication requirements.
- Automated documentation generation and publishing for +50 daily deployments, reducing documentation fixes by 60% through CI/CD, GitHub Actions and Kubernetes.
- Centralised administration portals and public documentation for the AI Gigafactory, improving onboarding and reducing manual queries for all internal teams and integrated external partners while streamlining team-stakeholder communication with React, TypeScript, Next.js, Docker and Docusaurus.
- Contributed to scaling a cloud-native AI computing platform to sustain data-centre loads of >10K events/s through idempotent OTLP log replay, Kubernetes network policies, end-to-end testing and AKS cluster lifecycle automation.

Support Engineer - Telefónica, BCNC Group
03/2023 - 01/2024 · Madrid, Spain

- Built and operated a centralised mock platform for global telco consent management and authentication, covering Node.js API, IdP and consent-management mocks for CAMARA APIs including SIM Swap and KYC. Supported 3 cross-functional projects with routine traffic of >10K requests/hour and peaks of 5K requests/minute, enabling integration testing without full environments, reducing QA costs and accelerating delivery.
- Designed and validated third-party REST API integrations and OAuth flows for partners within Open Gateway and CAMARA, translating API and identity requirements into mock behaviour, test collections and technical documentation; diagnosed integration issues with platform and API teams.
- Operated integration environments and, with Operations, investigated and resolved pre-production and production incidents in backend microservices, Apache Spark and Apache Kafka.
- Reduced support and maintenance work by 50% with a Python CLI, internal flow-management tools and CI/CD automation for imports, cleanup and application lifecycle, making CAMARA and Open Gateway integration testing reproducible.

Backend & Architecture R&D Lead - BISITE Research Group
09/2020 - 03/2023 · Salamanca, Spain

- Led the technical delivery of 10 SaaS platforms to production with a 5-person engineering team, including a platform that processed 60K events/hour at peak.
- Designed cloud-native R&D and AI platforms, including Apache Kafka-based ETL systems, Python and Node.js microservices, and internal tooling.

Full-stack Developer - BISITE Research Group
06/2019 - 09/2020 · Salamanca, Spain

- Maintained the DeepInt Python SDK and added authentication, documentation, tests and higher-level workflows for users of the information-ingestion and AI-analysis SaaS.
- Built research and partner platforms with Apache Kafka-based ETL and MEVN/MERN applications; DeepInt processed >100K events/hour.

Lecturer - University and professional programmes
01/2021 - present · Spain

- **University of Salamanca:** taught Generative AI Project Design, Smart Cities Asset Management, IoT architectures and introductory artificial intelligence across postgraduate programmes.
- **Sociedad de Promoción Económica de Gran Canaria (SPEGC):** taught introductory artificial intelligence, convolutional neural networks and recurrent neural networks in 2022.
- **Pontifical University of Salamanca:** taught Database Management and Fundamentals of Computer Science I and II in 2022.

Education

PhD in Computer Science and Artificial Intelligence - University of Salamanca
10/2022 - 07/2026 · Salamanca, Spain

- Grade: *Summa cum laude*.
- Thesis: *Trustworthy Distributed Architectures for Intelligent Information Processing*.
- Research on event-driven platforms with Kafka, Flink, NLP microservices, polyglot persistence, idempotent processing and privacy by design.
- Controlled experiments sustained 100K events/minute with 3 s p95 latency for real-time NLP processing.

MSc in Computer Engineering - University of Salamanca
09/2020 - 09/2022 · Salamanca, Spain

- GPA: 9.57/10; first in the 2020-2022 class.
- Thesis: *DeepTrends: Distributed Platform for Information Retrieval and Processing*, a distributed real-time NLP architecture with event-driven processing and polyglot persistence.

BSc in Computer Engineering - University of Salamanca
09/2016 - 07/2020 · Salamanca, Spain

- GPA: 9.55/10; first in the 2016-2020 class.
- Technical final project: *NovaTrends: Online Visualisation Platform for the Value of Technology Trends*, a distributed pipeline for online technology-trend analysis with NLP enrichment.

Achievements

- Third place, University of Salamanca Alumni Talent Award (04/2026).
- Best academic record award for the 2020-2022 MSc in Computer Engineering class, University of Salamanca (01/2023).
- Best academic record award for the 2016-2020 BSc in Computer Engineering class, University of Salamanca (07/2021).
- Third place in SEDEA's national ranking of engineering students (07/2020).